

Nishant Balepur

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RESEARCH SUMMARY

I am a computer science Ph.D. candidate at the University of Maryland, advised by Professor Jordan Boyd-Graber and Rachel Rudinger. I study Natural Language Processing, but apply tools from Machine Learning and Human-Computer Interaction. I develop and deploy collaborative AI systems that measurably help users achieve their goals, with a goal of improving user feedback, personalization, reasoning, factuality, and evaluations beyond just correctness and preferences.

EDUCATION

University of Maryland, College Park (UMD)

Ph.D. Computer Science; GPA: 4.00/4.00

College Park, MD

Aug 2023 - Present

Advisors: Jordan Boyd-Graber, Rachel Rudinger

Proposed Thesis: Teaching AI to Answer Questions with Reasoning that Actually Helps You [PDF]

Committee: Jordan Boyd-Graber, Rachel Rudinger, Shi Feng, Fumeng Yang, David Weintrop

New York University

Visiting Student

Host: Eunsol Choi

New York, NY

Aug 2025 - May 2026

University of Illinois at Urbana-Champaign (UIUC)

B.S. Computer Science; B.S. Statistics (Dual Degree); GPA: 4.00/4.00

Urbana, IL

Aug 2019 - May 2023

INDUSTRY RESEARCH EXPERIENCE

Allen Institute for AI (Ai2)

Research Scientist Intern: Asta (Self-Evolving Human-AI Interaction)

Seattle, WA

Sept 2026 - Present

Microsoft Research (MSR)

Research Intern: AI Interaction and Learning (Coding Agent Evaluation)

Outcome: One paper under review at ICLR'27

New York, NY

May 2026 - Aug 2026

Allen Institute for AI (Ai2)

Research Scientist Intern: Semantic Scholar (Personalized Deep Research)

Outcome: One ACL'26 paper, One paper under review, co-authored ICLR'26 oral, Prod demo

Seattle, WA

May 2025 - May 2026

Adobe

Research Scientist Intern: Document Intelligence (Multi-LLM QA)

Outcome: Paper at NAACL 2025

San Jose, CA

May 2024 - Aug 2024

Meta

Software Engineering Intern: Facebook Creators Well-being (Comment Safety)

Outcome: Intervention with 15% drop in negative interactions across Facebook's 1 billion users

Menlo Park, CA

May 2022 - Aug 2022

FELLOWSHIPS, GRANTS, AND SCHOLARSHIPS

NSF Graduate Research Fellowship Program (GRFP)

Funded proposal on NLP for information accessibility, awarding \$159,000 over three years

UMD

April 2023 - April 2026

Cohere for AI Research Grant (2x Awardee)

Funded proposals on LLM for Education and AI Safety, receiving full access to Cohere models

UMD

April 2024

Dean's Fellowship

Awarded the Dean's Fellowship for outstanding academic achievement

UMD

April 2023 - April 2025

Illinois Engineering Achievement Scholarship (2x Awardee)

Merit-based scholarship for undergraduate students

UIUC

June 2021, 2022

AWARDS

Certificate of Outstanding Achievement (2x)

Outstanding performance and exceptional contributions to the CS Graduate Program

UMD

April 2025, 2026

MASC-SLL 2025 Best Paper Award

Best paper award for our MCQA Position Paper by Penn State University

UMD

April 2025

MASC-SLL 2024 Best Paper Award	UMD
Best paper award for “Artifacts or Abduction” by Johns Hopkins University	May 2024
Outstanding Reviewer Award (2x Awardee)	UMD
Selected as one of 51 outstanding reviewers for the EMNLP conference out of 10,000+ reviewers	Nov 2024, 2025
Bronze Tablet	UIUC
Name inscribed in the university library for graduating in the top 3% of students in the class of 2023	May 2023
Computer Science Graduation with Highest Honors	UIUC
Recommended by the UIUC computer science department to graduate with highest honors	May 2023
C.W. Gear Outstanding Undergraduate Student	UIUC
Awarded to two undergraduate CS seniors (fourth year) for excellence in research and service	May 2022
James N. Snyder Memorial Award	UIUC
Awarded to three undergraduate CS juniors (third year) based on academic merit	May 2021
UIUC Datathon Champion	UIUC
Developed the best machine learning model for analyzing consumer behavior out of 150 UIUC students	April 2021
State Farm Programming Competition Finalist	UIUC
Top four finalist out of 50 national teams for solving data analysis questions	Nov 2020

CONFERENCE PUBLICATIONS

* denotes equal contribution, † denotes mentored student

20. Check The Scoreboard: An Analysis of Scoring Schemes on Multiple-Choice Evaluation [PDF]
EMNLP 2026
Nishant Balepur, Paiheng Xu, Wei Ai, Eunsol Choi, Rachel Rudinger, Jordan Boyd-Graber
19. VibeJam: An Open Platform for User Studies on Agentic Vibe Coding [PDF]
EMNLP 2026 (Demo), North East AI Agents Day (**Oral**)
Nishant Balepur, Connor Baumler, Valerie Chen, Eunsol Choi, Rachel Rudinger, Jordan Boyd-Graber
18. Language Models Don’t Know What You Want: Personalization in Deep Research Needs Real Users [PDF]
ACL 2026
Nishant Balepur, Sergey Feldman, ..., Joseph Chee Chang, Eunsol Choi, Jordan Boyd-Graber, Aakanksha Naik
Selected: We deploy a personalized Deep Research system and collect user feedback to uncover novel failures
17. Test-Time Reasoners Are Strategic Multiple-Choice Test-Takers [PDF]
ACL 2026
Nishant Balepur, Atrey Desai†, Rachel Rudinger
16. BenchMarker: An Education-Inspired Toolkit for Highlighting Flaws in Multiple-Choice Benchmarks [PDF]
ACL 2026
Nishant Balepur, Bhavya Rajasekaran†, Jane Oh†, Michael Xie†, Atrey Desai†, ..., Jordan Boyd-Graber
15. Measuring Users’ Mental Models of Speech Translation in Human-MT Collaboration [PDF]
ACL 2026, IWSLT 2026 (**Oral**)
Hyojung Han, **Nishant Balepur**, Jordan Boyd-Graber, Marine Carpuat
14. AstaBench: Rigorous Benchmarking of AI Agents with a Holistic Scientific Research Suite [PDF]
ICLR 2026 (**Oral**)
Jonathan Bragg*, Mike D’Arcy*, **Nishant Balepur**, ..., Yoav Goldberg, Ashish Sabharwal, Daniel S. Weld
13. A Good Plan is Hard to Find: Aligning Models with Preferences is Misaligned with What Helps Users [PDF]
EMNLP 2025
Nishant Balepur, Matthew Shu†, ..., Shi Feng, Fumeng Yang, Rachel Rudinger, Jordan Boyd-Graber
Selected: We show optimizing on user/LLM judgments and agents simulations cannot capture what helps users
12. Which of These Describes Multiple Choice Evaluations? A) Forced B) Flawed C) Fixable D) All of the Above [PDF]
ACL 2025 (**Oral, ACL Best Paper Nomination, Best Paper Award at MASC-SLL 2025**)
Nishant Balepur, Rachel Rudinger, Jordan Boyd-Graber
Selected: We expose flaws multiple-choice evaluations and design solutions from education to remedy them
11. Whose Boat Does it Float? Improving Personalization in Preference Tuning via Inferred User Personas [PDF]
ACL 2025
Nishant Balepur, Vishakh Padmakumar, Fumeng Yang, Shi Feng, Rachel Rudinger, Jordan Boyd-Graber

10. Reverse Question Answering: Can an LLM Write a Question so Hard (or Bad) that it Can't Answer? [PDF]
NAACL 2025, **Oral**
Nishant Balepur, Feng Gu, Abhilasha Ravichander, Shi Feng, Jordan Boyd-Graber, Rachel Rudinger
9. Moderating a Mixture of Document Speakers to Summarize Debatable Queries in Document Collections [PDF]
NAACL 2025, **Oral**
Nishant Balepur, Alexa Siu, Nedim Lipka, Franck Dernoncourt, Tong Sun, Jordan Boyd-Graber, Puneet Mathur
8. A Smart Mnemonic Sounds like Glue Tonic: Mixing LLMs with Student Feedback to Make Learning Stick [PDF]
EMNLP 2024
Nishant Balepur, Matthew Shu†, Alexander Hoyle, ..., Shi Feng, Seraphina Goldfarb-Tarrant, Jordan Boyd Graber
Selected: We optimize language models directly on student learning outcomes to improve mnemonic generation
7. KARL: Knowledge-Aware Retrieval and Representations aid Retention and Learning in Students [PDF]
EMNLP 2024
Matthew Shu*†, **Nishant Balepur***, Shi Feng*, Jordan Boyd-Graber
6. Plausibly Problematic Questions in Multiple-Choice Benchmarks for Commonsense Reasoning [PDF]
EMNLP 2024 (Findings)
Shramay Palta, **Nishant Balepur**, Peter Rankel, Sarah Wiegrefe, Marine Carpuat, Rachel Rudinger
5. Artifacts or Abduction: How Do LLMs Answer Multiple-Choice Questions Without the Question? [PDF]
ACL 2024, **Best Paper Award at MASC-SLL 2024**
Nishant Balepur, Abhilasha Ravichander, Rachel Rudinger
Selected: We discover LLMs score highly in MCQA benchmarks without the question, and analyze how
4. It's Not Easy Being Wrong: Large Language Models Struggle with Process of Elimination Reasoning [PDF]
ACL 2024 (Findings)
Nishant Balepur, Shramay Palta, Rachel Rudinger
3. Expository Text Generation: Imitate, Retrieve, Paraphrase [PDF]
EMNLP 2023, **On UIUC's AI Reading List for Qualifying Exams, On Stanford's Reading List for CS224V**
Nishant Balepur, Jie Huang, Kevin Chen-Chuan Chang
2. Text Fact Transfer [PDF]
EMNLP 2023
Nishant Balepur, Jie Huang, Kevin Chen-Chuan Chang
1. DynaMiTE: Discovering Explosive Topic Evolutions with User Guidance [PDF]
ACL 2023 (Findings)
Nishant Balepur*, Shivam Agarwal*, Karthik Ramanan, Susik Yoon, Diyi Yang, Jiawei Han

PREPRINTS AND TECHNICAL REPORTS

* denotes equal contribution, † denotes mentored student

6. Programming is Not Just Editing! The Importance of Code Understanding in Agents [PDF]
Under Review (ICLR)
Nishant Balepur, Kiran Tomlinson, Tobias Schnabel
5. (Im)Paired Programming: Coding Agents Improve Productivity but Harm Understanding [PDF]
Under Review (ARR)
Nishant Balepur, Connor Baumler, Valerie Chen, Eunsol Choi, Rachel Rudinger, Jordan Boyd-Graber
4. DRACULA: Hunting for the Actions Users Want Deep Research Agents to Execute [PDF]
Under Review (ARR)
Nishant Balepur, ..., Rachel Rudinger, Eunsol Choi, Jordan Boyd-Graber, Doug Downey, Aakanksha Naik
3. Quick, Create a Distractor! Evaluating LLM Distractors for Multiple-Choice Benchmarks [PDF]
Under Review (ARR)
Atrey Desai†, **Nishant Balepur**, Rachel Rudinger
2. Position: Cognitive Alignment for Human Empowerment [PDF]
Under Review (NeurIPS)
Yoonjoo Lee, Philippe Laban, **Nishant Balepur**, Hariharan Subramonyam, Vera Liao
1. The Prompt Report: A Systematic Survey of Prompting Techniques [PDF]
Technical Report, **Huggingface #1 Paper of the Day**
Sander Schulhoff*, Michael Ilie*, **Nishant Balepur**, ..., Shyamal Anadkat, Alexander Hoyle, Phillip Resnik

WORKSHOP PAPERS

* denotes equal contribution, † denotes mentored student

4. BenchMarkerUI: An Interactive System for Auditing and Repairing Multiple-Choice Benchmarks
HCOMP 2026
Hyunjin Jane Oh†, **Nishant Balepur**, Jordan Boyd-Graber
3. Can They Dixit? Yes they Can! Dixit as a Playground for Multimodal Language Model Capabilities [PDF]
Wordplay @ EMNLP 2025, **Spotlight**
Nishant Balepur*, Dang Nguyen*, Dayeon Ki*
2. Language Models Generate Multiple-Choice Questions with Artifacts [Poster]
MASC-SLL 2025
Atrey Desai†, **Nishant Balepur**, Rachel Rudinger
1. Is Your Large Language Model Knowledgeable or a Choices-Only Cheater? [PDF]
KnowLM @ ACL 2024
Nishant Balepur, Rachel Rudinger

INVITED TALKS

Understanding Understanding in Coding Agents Pangram	New York City August 2026
BenchMarker: A Toolkit for Highlighting Flaws in Multiple-Choice Benchmarks CAMEL-AI	Virtual July 2026
Towards Helpful AI Agents FigX Lab, University of Maryland	Virtual March 2026
Aligning Models with Preferences is Misaligned with What Helps Users ML2 Mixer, New York University	New York, New York November 2025
Which of These Best Describes Multiple-Choice Evaluation with LLMs? Google Translate Reading Group	Virtual March 2025
Aligning LLMs to Answer Questions with Reasoning That Actually Helps You Imperial College, London	London, U.K. March 2025

STUDENT MENTEES

Matthew Shu (B.S. Yale → M.S. Yale → ML @ Brain Co.), 2023-2025, LLMs for Education, Alignment
Two papers at EMNLP 2024 (long, main); one paper at EMNLP 2025 (long, main)

Jerry He (HS Student → B.S. GTech), 2024-2025, LLMs for Education
Submitted Project on LLMs+Education for Marriott's Ridge High School Gifted and Talented Research Program

Michael Xie (M.S. UMD → Ph.D. JHU), 2025-2026, Personalized LLMs for Education, LLM Evaluation
One paper at ACL 2026, One paper in preparation

Atrey Desai (B.S. UMD), 2024-Present, LLM Evaluation
One paper at MASC-SLL 2025, Two papers at ACL 2026, One paper under review at EMNLP

Jane Oh (B.S. UMD), 2025-Present, LLM Evaluation, Human-AI Collaboration
One paper at ACL 2026, One paper in preparation

Bhavya Rajasekaran (B.S. UMD), 2025, LLM Evaluation
One paper at ACL 2026

SERVICE, EXTRACURRICULARS, AND TEACHING

Workshop Organizer Cognitive Alignment: Aligning AI with Human Cognitive Processes (Under Review) Scholarly Document Processing Workshop – Shared Task (Under Review) The Science of Personas and LLM Personalization (Under Review)	UMD 2026
Conference Reviewer Reviewer for: *ACL/ARR 2023–Present, IEEE TASLP 2024, TrustNLP 2024, SRW 2025	UMD 2022–Present

Great Reviewer Nomination: Apr, June (#6), Aug (#4), Oct (#2), Dec 2024, Feb 2025 ARR
Outstanding Reviewer Award: EMNLP 2024, EMNLP 2025
Ranked #4 Great Reviewer Overall: ARR

Conference Volunteer

ACL 2023, NAACL 2025

UMD
2023, 2025

Winter Storm LLM Workshop

Led a five-day workshop on LLM research for non-CS students in the language science center

UMD
Jan 2023

Visiting Student Day Volunteer

Volunteer for UMD's visiting student day for prospective graduate students

UMD
Mar 2024, 2025

Undergraduate TA for Linear Algebra with Computation Methods

Led weekly office hours for 100+ students, created coding HW, and wrote outreach blogs

UIUC
Aug 2020 - May 2023

Computer Science and Statistics Student Ambassador

Mentor of ten incoming students and volunteer for computer science and statistics events

UIUC
Aug 2022 - May 2023

SIGNLL

President of Special Interest Group for Natural Language Learning

UIUC
Aug 2020 - May 2021

Project: Code

Co-founder of student organization to help new students build computer science projects

UIUC
Aug 2019 - May 2021